



**Generation Interconnection
System Impact Study Report
for
Queue Project AF1-245
HUDSON 230 KV
80 MW Capacity / 200 MW Energy**

August 2020

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1 Introduction

This System Impact Study has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the System Impact Study Agreement between the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is PSEG.

2 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

An Interconnection Customer with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM Manual 14D for additional information.

3 General

The Interconnection Customer (IC), **Baywell Energy, LLC**, has proposed a Storage generating facility located in Hudson County, New Jersey. The installed facilities will have a total capability of 200 MW with 80 MW of this output being recognized by PJM as Capacity. The proposed in-service date for this project is May 31, 2023. This study does not imply a TO commitment to this in-service date.

Queue Number	AF1-245
Project Name	HUDSON 230 KV
State	New Jersey
County	Hudson
Transmission Owner	PSEG
MFO	200
MWE	200
MWC	80
Fuel	Storage
Basecase Study Year	2023

Any new service customers who can feasibly be commercially operable prior to June 1st of the basecase study year are required to request interim deliverability analysis.

4 Point of Interconnection

AF1-245 will interconnect with the PSEG on transmission system at the Hudson 230 kV substation. The Point of Interconnection will be located at the PSEG monopole one (1) span outside of the IC's collector substation

5 Cost Summary

The AF1-245 project will be responsible for the following costs:

Description	Total Cost
Total Physical Interconnection Costs	\$5,780,000
Allocation towards System Network Upgrade Costs*	\$0
Total Costs	\$5,780,000

*As your project progresses through the study process and other projects modify their request or withdraw, then your cost allocation could change.

This cost excludes a Federal Income Tax Gross Up charges. This tax may or may not be charged based on whether this project meets the eligibility requirements of IRS Notice 88-129. If at a future date it is determined that the

Federal Income Tax Gross charge is required, the Transmission Owner shall be reimbursed by the Interconnection Customer for such taxes.

Note 1: PJM Open Access Transmission Tariff (OATT) section 217.3A outline cost allocation rules. The rules are further clarified in PJM Manual 14A Attachment B. The allocation of costs for a network upgrade will start with the first Queue project to cause the need for the upgrade. Later queue projects will receive cost allocation contingent on their contribution to the violation and are allocated to the queues that have not closed less than 5 years following the execution of the first Interconnection Service Agreement which identifies the need for this upgrade.

Note 2: For customers with System Reinforcements listed: If your project's present cost allocation to a System Reinforcement indicates \$0, please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your project's present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

6 Transmission Owner Scope of Work

The total physical interconnection costs is given in the table below:

Description	Total Cost
Attachment Facilities	\$2,420,000
Install one (1) single circuit monopole and associated conductor to connect into an existing bus position, relays and revenue grade metering.	
Direct Connection Network Upgrades	\$0
None	
Non-Direct Connection Network Upgrades	\$3,360,000
Install two (2) double circuit monopoles and reconfigure two (2) transmission circuits and associated relays.	
Total Physical Interconnection Costs	\$5,780,000

Assumptions:

It is assumed that the existing structures can accommodate this queue request. No structural analysis was performed on the existing proposed to be used.

7 Schedule

The estimated schedule to complete the Transmission Owner scope of work is **24 months**.

8 Revenue Metering and SCADA Requirements

8.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

8.2 Interconnected Transmission Owner Requirements

The IC will be required to comply with all Interconnected Transmission Owner's revenue metering requirements for generation interconnection customers located at the following link:

<http://www.pjm.com/planning/design-engineering/to-tech-standards/>

9 Summer Peak Analysis

The Queue Project AF1-245 was evaluated as a 200 MW (Capacity 80.0 MW) injection at the **Hudson 230 kV substation** in the PSEG area. Project AF1-245 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AF1-245 was studied with a commercial probability of 100.0 %. Potential network impacts were as follows:

9.1 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

9.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

9.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

9.4 Steady-State Voltage Requirements

To be determined

9.5 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
49627161	217134	BAYWAY	345	PSE&G	216908	BAYONNE	345	PSE&G	1	Base Case	operation	670.0	99.35	102.17	AC	23.1

9.6 System Reinforcements

None. See section 10.6 for Light Load Analysis System Reinforcements.

10 Light Load Analysis

The Queue Project AF1-245 was evaluated as a 200 MW injection at the Hudson 230 kV substation in the PSEG area. Project AF1-245 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AF1-245 was studied with a commercial probability of 100 %. Potential network impacts were as follows:

10.1 Generation Deliverability

(Single or N-1 contingencies)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
71758356	213922	RICHMOND	230.0	PECO	214012	WANEETA3	230.0	PECO	1	PJM500_PS_P1-2_5015	single	1180.0	99.79	100.72	AC	11.14

10.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies)

None

10.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

10.4 Steady-State Voltage Requirements

None

10.5 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
49626152	213922	RICHMOND	230.0	PECO	214012	WANEETA3	230.0	PECO	1	PJM500_PS_P1-2_5015	operation	1180.0	99.43	100.21	AC	11.14

10.6 System Reinforcements

ID	Index	Facility	Upgrade Description	Cost	Cost Allocated to AF1-245	Upgrade Number
71758356	1	RICHMOND 230.0 kV - WANEETA3 230.0 kV Ckt 1	Project ID : N6477 (pe010) Description : Add a second pair of underground cables in parallel with the existing underground portion of the Richmond - Waneeta 230 kV line (Does not include permitting) Type : FAC Total Cost : \$19,000,000 Time Estimate : 84.0 Months Ratings : 1026.0/1247.0/1337.0 Notes: This upgrade is driven by a prior queue (AE2-021 as of 8/31/2020) for a different constraint. Per PJM cost allocation rules, Queue Project AF1-245 presently does not receive cost allocation for this upgrade.	\$19,000,000	\$0	N6477
			TOTAL COST	\$19,000,000	\$0	

Note:

For customers with System Reinforcements listed:

If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc., the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your project's present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

10.7 Flow Gate Details

The following indices contain additional information about each facility presented in the body of the report. For each index, a description of the flowgate and its contingency was included for convenience. The intent of the indices is to provide more details on which projects/generators have contributions to the flowgate in question. All New Service Queue Requests, through the end of the Queue under study, that are contributors to a flowgate will be listed in the indices. Please note that there may be contributors that are subsequently queued after the queue under study that are not listed in the indices. Although this information is not used "as is" for cost allocation purposes, it can be used to gage the impact of other projects/generators. It should be noted the project/generator MW contributions presented in the body of the report are Full MW Impact contributions which are also noted in the indices column named "Full MW Impact", whereas the loading percentages reported in the body of the report, take into consideration the PJM Generator Deliverability Test rules such as commercial probability of each project as well as the ramping impact of "Adder" contributions. The MW Impact found and used in the analysis is shown in the indices column named "Gendeliv MW Impact".

10.8 Index 1

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
71758356	213922	RICHMOND	PECO	214012	WANEETA3	PECO	1	PJM500_PS_P1-2_5015	single	1180.0	99.79	100.72	AC	11.14

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206325	280 C GEN (Deactivation : 17/09/2018)	33.0299	80/20	33.0299
936192	AD2-025 E	0.1311	80/20	0.1311
937512	AD2-210 E (Suspended)	0.1032	80/20	0.1032
937563	AD2-077 BAT	12.2380	80/20	12.2380
938231	AE1-037 FTIR (Withdrawn : 01/24/2020)	70.8192	80/20	70.8192
938281	AE1-041 C	0.0001	80/20	0.0001
938282	AE1-041 E	0.0346	80/20	0.0346
938421	AE1-061 C	0.6771	80/20	0.6771
938422	AE1-061 E	0.6771	80/20	0.6771
938431	AE1-062 C	2.5750	80/20	2.5750
938432	AE1-062 E	2.5750	80/20	2.5750
938781	AE1-104 C O1	21.6500	80/20	21.6500
938782	AE1-104 E O1	55.3912	80/20	55.3912
938871	AE1-115 C	2.5947	80/20	2.5947
938872	AE1-115 E	2.5947	80/20	2.5947
939301	AE1-161 C	5.0580	80/20	5.0580
939302	AE1-161 E	7.5870	80/20	7.5870
939981	AE1-238 C	19.4094	80/20	19.4094
939982	AE1-238 E	50.9820	80/20	50.9820
940161	AE2-000 C O1	26.9528	80/20	26.9528

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
940162	AE2-000 E O1	68.9631	80/20	68.9631
940361	AE2-020 C	18.3886	80/20	18.3886
940362	AE2-020 E	86.0967	80/20	86.0967
940371	AE2-021 C	18.3886	80/20	18.3886
940372	AE2-021 E	86.0967	80/20	86.0967
940381	AE2-022 C	10.7267	80/20	10.7267
940382	AE2-022 E	50.2231	80/20	50.2231
940401	AE2-024 C O1	15.1231	80/20	15.1231
940402	AE2-024 E O1	70.8048	80/20	70.8048
940411	AE2-025 C	7.6341	80/20	7.6341
940412	AE2-025 E	35.7390	80/20	35.7390
942101	AE2-222 C O1	15.0346	80/20	15.0346
942102	AE2-222 E O1	37.7678	80/20	37.7678
942381	AE2-251 C	58.2547	80/20	58.2547
942382	AE2-251 E	149.0573	80/20	149.0573
943561	AF1-027	0.3274	80/20	0.3274
944332	AF1-101 E O1	67.6113	80/20	67.6113
944441	AF1-109	1.1968	80/20	1.1968
944611	AF1-126 C O1	9.8680	80/20	9.8680
944612	AF1-126 E O1	39.4720	80/20	39.4720
944951	AF1-160 C	2.5750	80/20	2.5750
944952	AF1-160 E	2.5750	80/20	2.5750
945201	AF1-185 1	0.3650	80/20	0.3650
945211	AF1-185 2	0.0811	80/20	0.0811
945431	AF1-208 C O1	0.0002	80/20	0.0002
945432	AF1-208 E O1	2.4902	80/20	2.4902
945571	AF1-222 C	12.0985	80/20	12.0985
945572	AF1-222 E	31.8961	80/20	31.8961
945721	AF1-237 C	13.1050	80/20	13.1050
945722	AF1-237 E	0.0001	80/20	0.0001
945731	AF1-238 C	12.1089	80/20	12.1089
945732	AF1-238 E	18.1634	80/20	18.1634
945741	AF1-239 C	2.8492	80/20	2.8492
945742	AF1-239 E	4.2737	80/20	4.2737
945801	AF1-245 C O1	4.4576	80/20	4.4576
945802	AF1-245 E O1	6.6864	80/20	6.6864

10.9 Queue Dependencies

The Queue Projects below are listed in one or more indices for the overloads identified in your report. These projects contribute to the loading of the overloaded facilities identified in your report. The percent overload of a facility and cost allocation you may have towards a particular reinforcement could vary depending on the action of these earlier projects. The status of each project at the time of the analysis is presented in the table. This list may change as earlier projects withdraw or modify their requests.

Queue Number	Project Name	Status
AD2-025	Hillsborough 13 kV	Engineering and Procurement
AD2-077	Buxmont 69 kV	Active
AD2-210	Cedar Knolls 12.5 kV	Suspended
AE1-037	Deans 500kV	Withdrawn
AE1-041	Edison 4kV	In Service
AE1-061	Minotola 12 kV	Active
AE1-062	Silver Lake 69 kV	Active
AE1-104	BL England 138 kV	Active
AE1-115	Churchtown 69 kV	Active
AE1-161	Landis 138 kV	Active
AE1-238	Oceanview Wind 230 kV	Active
AE2-000	N/A	N/A
AE2-020	Cardiff 230 kV I	Active
AE2-021	Cardiff 230 kV II	Active
AE2-022	Cardiff 230 kV III	Active
AE2-024	Larrabee 230 kV I	Active
AE2-025	Larrabee 230 kV II	Active
AE2-222	Higbee 69 kV	Active
AE2-251	Cardiff 230 kV	Active
AF1-027	Plumsted 537 Energy Storage (CIRs)	Active
AF1-101	Oyster Creek 230 kV III	Active
AF1-109	Pleasant Valley 230 kV	Active
AF1-126	Cookstown 230 kV	Active
AF1-160	Silver Lake 69 kV	Active
AF1-185	Sayreville 1-2-3 230 kV	In Service
AF1-208	Quinton-Roadstown 69 kV	Active
AF1-222	Oceanview Wind 2 230 kV	Active
AF1-237	Mercer 230 kV	Active
AF1-238	Sherman Ave. 69 kV	Active
AF1-239	Sherman Ave-Vineland 69 kV	Active
AF1-245	Hudson 230 kV	Active

10.10 Contingency Descriptions

Contingency Name	Contingency Definition
PJM500_PS_P1-2_5015	CONTINGENCY 'PJM500_PS_P1-2_5015' TRIP LINE FROM BUS 200027 TO BUS 200029 CKT 1 END /* HOPE CREEK TO RED LION /* HOPE CREEK TO RED LION

11 Short Circuit Analysis

The following Breakers are overdutied

None

12 Stability and Reactive Power

(Summary of the VAR requirements based upon the results of the dynamic studies)

To be determined in the Facilities Study Phase. Additional System reinforcements may be identified to address stability concerns.

13 Affected Systems

13.1 NYISO

NYISO Impacts to be determined during later study phases (as applicable).

14 Attachment 1: One Line Diagram

